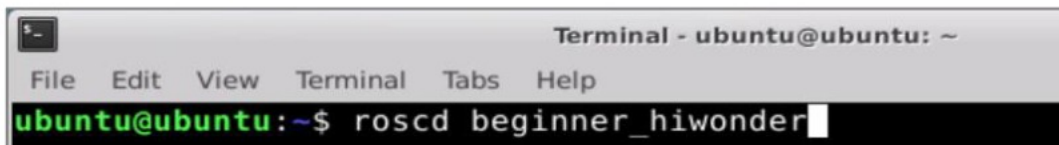


## 第 5 课 编写简单的发布者 Publisher

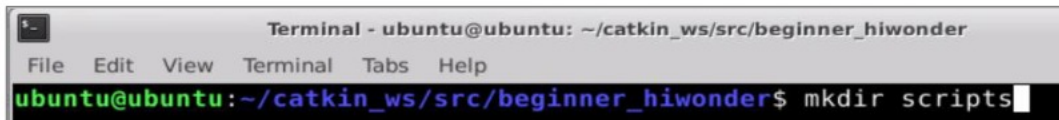
本节以创建一个velocity\_publisher.py的（发布者）节点为例进行讲解。

- 1) 输入指令“`roscd beginner_hiwonder`”，回车。进入beginner\_hiwonder软件包。



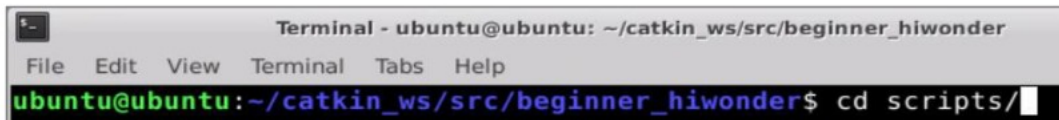
```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd beginner_hiwonder
```

- 2) 输入指令“`mkdir scripts`”，回车。新建一个scripts目录存放Python脚本。



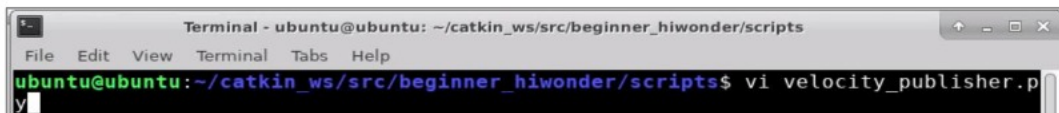
```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir scripts
```

- 3) 输入指令“`cd scripts/`”，回车。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ cd scripts/
```

- 4) 输入指令“`vi velocity_publisher.py`”编辑程序，复制下面程序。如需修改，再按下“i”即可修改。修改完成，按下“Esc”，输入“: wq”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi velocity_publisher.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将发布turtle1/cmd_vel话题，消息类型geometry_msgs::Twist

import rospy

from geometry_msgs.msg import Twist
```

```

def velocity_publisher():

    # ROS节点初始化

    rospy.init_node('velocity_publisher', anonymous=True)

    # 创建一个 Publisher，发布名为 /turtle1/cmd_vel 的 topic，消息类型为
    geometry_msgs::Twist，队列长度10

    turtle_vel_pub = rospy.Publisher('/turtle1/cmd_vel', Twist, queue_size=10)

    #设置循环的频率

    rate = rospy.Rate(10)

    while not rospy.is_shutdown():

        # 初始化geometry_msgs::Twist类型的消息

        vel_msg = Twist()

        vel_msg.linear.x = 0.5

        vel_msg.angular.z = 0.2

        # 发布消息

        turtle_vel_pub.publish(vel_msg)

        rospy.loginfo("Publsh turtle velocity command[%0.2f m/s, %0.2f rad/s]",

            vel_msg.linear.x, vel_msg.angular.z)

```



```
# 按照循环频率延时

rate.sleep()

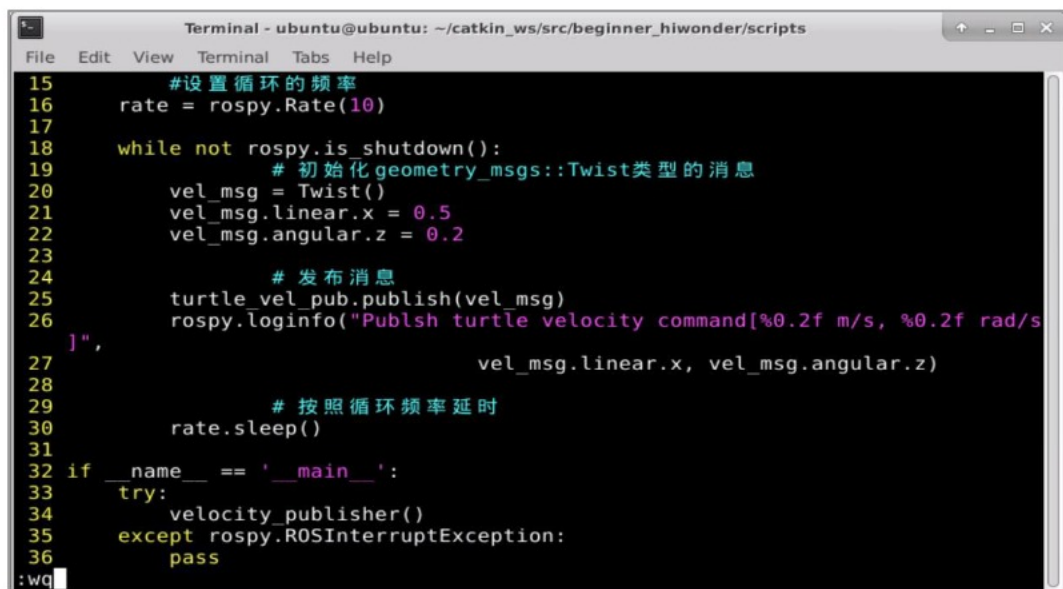
if __name__ == '__main__':

    try:

        velocity_publisher()

    except rospy.ROSInterruptException:

        pass
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
15 #设置循环的频率
16 rate = rospy.Rate(10)
17
18 while not rospy.is_shutdown():
19     # 初始化geometry_msgs::Twist类型的消息
20     vel_msg = Twist()
21     vel_msg.linear.x = 0.5
22     vel_msg.angular.z = 0.2
23
24     # 发布消息
25     turtle_vel_pub.publish(vel_msg)
26     rospy.loginfo("Publish turtle velocity command[%0.2f m/s, %0.2f rad/s",
27         vel_msg.linear.x, vel_msg.angular.z)
28
29     # 按照循环频率延时
30     rate.sleep()
31
32 if __name__ == '__main__':
33     try:
34         velocity_publisher()
35     except rospy.ROSInterruptException:
36         pass
:wq
```

5) 输入指令“**chmod +x velocity\_publisher.py**”回车，为保存的velocity\_publisher.py 赋予可执行权限。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x velocity_publisher.py
```

到此，发布者编辑完成。